

Chen-Han Lin

✉ allenchenhan99@gmail.com ☎ +886 978 261 955 📁 Portfolio in LinkedIn 🌐 GitHub

Education

National Yang Ming Chiao Tung University, College of Computer Science Sep 2023 – Jul 2026
M.S. in Data Science and Engineering

- **Coursework:** Machine Learning, Deep Learning, Data Science Project, Biostatistics

National Cheng Kung University, College of Science Sep 2018 – Jun 2022
B.S. in Mathematics

- **Coursework:** Probability and Statistics, Fourier Analysis, Data Structures and Algorithms

Master's Thesis

Model Enhancement and Implementation of Human Pose Estimation Based on the Mamba State Space Architecture

- Advisor: Prof. Chin-Tien Wu
- Integrated Mamba sequence blocks into a Video Pose Transformer for GPU-accelerated monocular 2D-to-3D pose estimation; reduced inference latency and training time through CUDA optimization, parallel data pipelines, and modular integration.

Internship Experience

Whale Force Holdings Ltd. Mar 2026 – Present
Data Scientist Intern

Systematic Strategy Research Built a backtesting engine for investment hypotheses, evaluating benchmark-relative hit rates, Sharpe ratios, and Monte Carlo robustness.

Financial Data Infrastructure Integrated automated financial-statement segmentation and parsing into downstream research pipelines.

Institutional Portfolio Research Developed a 13F-based clustering framework to identify investor segments with superior forward benchmark-relative performance.

Projects

Low-Rank Adaptation and Combined Parameter-Efficient Tuning for Large Models [Poster Link](#) 

- Investigated the integration of parameter-efficient fine-tuning (PEFT) methods such as LoRA, BitFit, Adapter, and Prefix-tuning across both vision and NLP tasks.
- Demonstrated that **LoRA + Adapter** achieved a **+3.6% improvement in accuracy** on vision classification tasks (from 0.8775 to 0.9134), with only **1.6% trainable parameters**.
- Showed that combining LoRA and BitFit improved NLP accuracy on RoBERTa-base from 0.931 to **0.945** (SST-2), confirming the complementary nature of these PEFT methods.

Data Science Project: Pima Indian Diabetes Dataset [Report Link](#) 

- Built classification pipelines across logistic regression, decision trees, random forests, and XGBoost, combining preprocessing, domain-specific feature engineering, stratified cross-validation, and grid search.
- Evaluated models with ROC-AUC, precision, recall, F1-score, and calibration curves, achieving **Accuracy = 83.6%**, **ROC-AUC = 91.2%**, **Precision = 84.5%**, **Recall = 88.2%**, and **F1 = 86.3%** on the PIMA Diabetes dataset.

Game Theory Applied to Darts Strategies

- Modeled checkout and target-selection decisions using professional match data, quantifying state-dependent risk-reward trade-offs to derive optimal decision rules across match scenarios.

Selected Honors & Competitions

TSMC IT CareerHack

Feb 2026

1st Place ranking

- Built a Docker-isolated refactoring platform with LangChain Deep Agents, FastAPI, React, and MongoDB, automating iterative analysis, refactoring, testing, and quality evaluation across three legacy repositories.

WorldQuant International Quant Championship

Mar 2026 - Jun 2026

4th Place in Taiwan

- Built a closed-loop alpha research engine that encoded backtest results as graph constraints, pruned redundant hypotheses, and re-ranked signals by marginal portfolio contribution.
- Evaluated 5,164 alpha expressions, with 1,023 passing all seven platform checks, and helped construct a 52-alpha portfolio that finished 4th in Taiwan.

TSMC IT CareerHack

Jan 2024

4th Place ranking

- Fine-tuned LLaVA with LoRA for task-specific visual question answering, supported by targeted data augmentation and noise filtering.

TSMC Intelligent Manufacturing Workshop

March 2025

Top-Performing Team, Scheduling Optimization

- Formulated the problem as a graph-based job-shop optimization, converting locations to nodes and distances to weighted edges, and applied Integer Linear Programming (ILP) to obtain the optimal schedule.

AI CUP 2024 Image Data Generation for UAVs

May 2024 - June 2024

Honorable Award

- Built a mask-conditioned image-generation pipeline with targeted augmentation and hyperparameter tuning, evaluated using Fréchet Inception Distance (FID).

AI GO Skyrocketed Stocks Forecasting

Mar 2025 – Apr 2025

- Built and tuned XGBoost, Random Forest, and TabNet models for stock-movement prediction using feature selection and tabular feature interactions.

Technical Skills

Programming, Data & Machine Learning:

- Python, C, C++, SQL, Bash; NumPy, pandas, scikit-learn, PyTorch, Transformers, TabNet, and Mamba
- Data preprocessing, feature engineering, ETL pipelines, and reproducible experimentation

Quantitative Research:

- Alpha research, backtesting, Monte Carlo simulation, portfolio and benchmark analysis, risk-adjusted performance evaluation, and statistical modeling
- Mathematical optimization using Integer Linear Programming and graph-based optimization methods

AI Engineering & Agentic Workflows:

- Coding agents, agent skills and harnesses, tool use, repository-level code generation, refactoring, debugging, and automated test generation
- Context engineering, task decomposition, workflow orchestration, iterative execution, agent evaluation, and human-in-the-loop validation

Software Engineering & Infrastructure:

- Linux, Git, Docker, Google Cloud Platform (GCP), Amazon Web Services (AWS), and MATLAB
- Modular system design, version control, debugging, performance profiling, and Linux-based data pipelines